

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{2}{27} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{36} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_d^{\mathbf{P}}] \right. \\ & \quad \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{12} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{8}{45} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{9} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{5}{24} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{4}{45} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{27} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{72} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{27} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{5}{9} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{32}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{9} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{6} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{8}{45} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{12} g_1^4 LF_{2,1,0}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{2,2,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{6} g_1^4 LF_{3,1,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{4,1,-2}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{12} g_1^4 LF_{2,1,0}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{2,2,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{6} g_1^4 LF_{3,1,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^4 LF_{4,1,-2}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^4 LF_{2,1,0}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{2,2,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{24} g_1^4 LF_{3,1,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{4,1,-2}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^4 LF_{2,1,0}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{2,2,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{24} g_1^4 LF_{3,1,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{48} g_1^4 LF_{4,1,-2}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{16} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{8} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{16} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{8} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{16} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{3} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} LF_{2,1,0}[m_e^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} + \frac{1}{6} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} LF_{2,2,-1}[m_e^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} + \\ & \quad \frac{1}{6} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} LF_{3,1,-1}[m_e^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} - \frac{1}{6} g_1^4 LF_{2,1,0}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{12} g_1^4 LF_{3,1,-1}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{6} g_1^4 LF_{2,1,0}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{12} g_1^4 LF_{3,1,-1}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} LF_{3,1,0}[m_l^{\mathbf{P}}, m_e^{\mathbf{r}}] + \\ & \quad \frac{3}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} LF_{4,1,-1}[m_l^{\mathbf{P}}, m_e^{\mathbf{r}}] - \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} LF_{5,1,-2}[m_l^{\mathbf{P}}, m_e^{\mathbf{r}}] - \\ & \quad \frac{1}{6} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{2,1,0}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{12} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{2,2,-1}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} + \\ & \quad \frac{1}{12} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{3,1,-1}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{24} g_1^4 LF_{2,1,0}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^4 LF_{3,1,-1}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{8} g_1^2 g_2^2 LF_{2,1,0}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{16} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^4 LF_{2,1,0}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{48} g_1^4 LF_{3,1,-1}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{8} g_1^2 g_2^2 LF_{2,1,0}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{16} g_1^2 g_2^2 LF_{3,1,-1}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{3}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_d^{\text{pr}} a_d^{\text{pr}} LF_{3,1,0}[m_q^{\mathbf{P}}, m_d^{\mathbf{r}}] + \\ & \quad \frac{9}{2} \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_d^{\text{pr}} a_d^{\text{pr}} LF_{4,1,-1}[m_q^{\mathbf{P}}, m_d^{\mathbf{r}}] - 3 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{a}_d^{\text{pr}} a_d^{\text{pr}} LF_{5,1,-2}[m_q^{\mathbf{P}}, m_d^{\mathbf{r}}] - \\ & \quad \frac{3}{2} \tilde{\mu}^2 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{y}_u^{\text{pr}} y_u^{\text{pr}} LF_{3,1,0}[m_u^{\mathbf{r}}, m_q^{\mathbf{P}}] + \frac{9}{2} \tilde{\mu}^2 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{y}_u^{\text{pr}} y_u^{\text{pr}} LF_{4,1,-1}[m_u^{\mathbf{r}}, m_q^{\mathbf{P}}] - \\ & \quad 3 \tilde{\mu}^2 \overline{y}_e^{i2i3} y_e^{i1i4} \overline{y}_u^{\text{pr}} y_u^{\text{pr}} LF_{5,1,-2}[m_u^{\mathbf{r}}, m_q^{\mathbf{P}}] + g_1^2 \overline{y}_e^{i2i3} y_e^{i1i4} LF_{3,1,-1}[\tilde{\mu}, m_1] - \\ & \quad 2 g_1^2 \overline{y$$